Yunsung Chung

PhD Candidate in Computer Science

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Professional Summary

Researcher in longitudinal EHR modeling, uncertainty-aware clinical AI, wearable biosignals, medical imaging, and clinically aligned generative models.

Education

PhD **Tulane University**, Computer Science — New Orleans, LA, USA
- Anticipated graduation: May 2027

Aug 2022 – present

MS **The Australian National University**, Computer Science — Canberra, Australia
July 2019 – July 2021

BS **Dongguk University**, Electronics, Information Communication Engineering — Seoul, South Korea
Mar 2012 – Feb 2018

Experience

Samsung Research America, Research Intern — Mountain View, CA, USA
Designed algorithms for predicting clinical biomarkers from large-scale wearable PPG data under sparse and noisy labels.
May 2025 – Aug 2025
4 months

TRIAD Lab, Tulane University, Research Assistant — New Orleans, LA, USA
Lead projects on long-horizon EHR modeling, offline-RL and OPE benchmarking, synthetic medical data, cardiac imaging, and wearable signal modeling.
Aug 2022 – present
4 years 4 months

Airport Railroad Express, Hardware Engineer — Incheon, South Korea
Nov 2017 – May 2019
1 year 7 months

Publications

EHRDyn: A Benchmark for Offline Reinforcement Learning from EHRs with Complementary Simulators — 2027
Yunsung Chung, Han Feng, Nassir Marrouche, Jihun Hamm
(KDD)

Reclassifying Single Lead Sinus Rhythm ECGs for Improved AF Recurrence Detection With Deep Learning — 2026
Yingshuo Liu, Chanho Lim, Ghassan Bidaoui, Christian Massad, Yunsung Chung, et al.
doi.org/10.1161/CIRCEP.125.014659 (Circulation: Arrhythmia and Electrophysiology, e014659)

Intervention-Aware Clinical World Model for Post–Op Outcome Forecasting in Cardiology — 2026
Yunsung Chung, Yingshuo Liu, Abboud F. Hassan, Han Feng, Mary M. Maleckar, Nassir Marrouche, Jihun Hamm
(MICCAI Workshop on Medical World Models (MWM), oral presentation)

Weighted Temporal Decay Loss for Learning Wearable PPG Data with Sparse Clinical Labels 2026
Yunsung Chung, Keum San Chun, Migyeong Gwak, Han Feng, Yingshuo Liu, Chanho Lim, Viswam Nathan, Nassir Marrouche, Sharanya Arcot Desai
(ICASSP)

SOFA: Deep Learning Framework for Simulating and Optimizing Atrial Fibrillation Ablation 2025
Yunsung Chung, Chanho Lim, Ghassan Bidaoui, Christian Massad, Nassir Marrouche, Jihun Hamm
(MICCAI)

SoK: Can Synthetic Images Replace Real Data? A Survey of Utility and Privacy of Synthetic Image Generation 2025
Yunsung Chung, Yunbei Zhang, Nassir Marrouche, Jihun Hamm
(USENIX Security)

Bridging the Gap: Synthetic Data Augmentation through Inversion and Distribution Matching for Few-Shot Learning 2024
Yunsung Chung, Janet Wang, Jihun Hamm
(AI-ML Systems)

From Majority to Minority: A Diffusion-Based Augmentation for Underrepresented Groups in Skin Lesion Analysis 2024
Janet Wang, Yunsung Chung, Zhengming Ding, Jihun Hamm
(MICCAI Workshop)

FBA-Net: Foreground and Background Aware Contrastive Learning for Semi-Supervised Atrium Segmentation 2023
Yunsung Chung, Chanho Lim, Chao Huang, Nassir Marrouche, Jihun Hamm
(MICCAI Workshop)

CRAFT: Clinical Reward-Aligned Finetuning for Medical Image Synthesis 2027
Yunsung Chung, Alex El Darzi, Carlo El Khoury, Han Feng, Nassir Marrouche, Jihun Hamm
(Under review at ICLR)

EHRGym: A Diagnostic Benchmark for Offline Reinforcement Learning 2027
Yunsung Chung, Han Feng, Nassir Marrouche, Jihun Hamm
(Under review at ICLR)

Deep Learning-Based Translation from Wearable ECG to PPG for PPG-Only Cardiac Rhythm Classification 2026
Yingshuo Liu, Alex El Darzi, Yunsung Chung, et al.
(Under review at npj Digital Medicine)

Awards

FIRA RoboWorld Cup - 1st Place in Basketball 2016

International Robot Contest - 2nd Place in Marathon 2016

First-Class Academic Excellence Scholarship, Dongguk University 2012

Skills

Machine Learning
Advanced; PyTorch, MONAI, diffusion models, synthetic data, LoRA, preference optimization

Medical AI
Advanced; Cardiac MRI, ECG/PPG modeling, biomarker prediction, privacy-utility analysis

Service

Reviewer: ICASSP 2027; NeurIPS 2026; ICML 2026; MICCAI 2024–2026; IEEE TCSVT, 2023.

Languages

Korean
Native

English
Fluent